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Aeration promotes Proteobacteria over Firmicutes in macerated food waste, resulting in superior anaerobic digestion efficiency

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Abstract

Aeration is a common pretreatment method to enhance biogas production via anaerobic digestion of waste organic feedstocks such as unused food. While impacts on downstream anaerobic digestion have been intensively investigated, the consequence of aeration on the microbial community in food waste has not been characterized. Food waste has a low pH resulting from the dominance of lactic acid bacteria within the Firmicutes phylum. This excludes other phylotypes with a higher potential to hydrolyse complex biopolymers in food waste. In this study, we reveal that aeration of macerated food waste results in a dramatic shift away from Firmicutes towards dominance of Proteobacteria that are better known for extracellular enzyme production. Given that hydrolysis is the rate limiting step in anaerobic digestion, this explains why aeration improves the efficiency of biogas production from food waste. The discovery that Proteobacteria dominate microbial communities in aerated food waste opens up opportunities to manipulate extracellular enzyme production through gene expression mechanisms common among Proteobacteria such as quorum sensing.

Keywords: food waste; anaerobic digestion; pretreatment; microbial ecology

Introduction

Pretreating food waste via physical, chemical, or biological methods enhances the efficiency of subsequent anaerobic digestion carried out by microorganisms (Suhartini et al. 2022). Despite efforts to develop pretreatment methods, little attention has been given to the microbiology of food waste prior to or during pretreatment, which logically influences the effectiveness of pretreatments that exploit microflora indigenous to food waste.

Lactic acid bacteria belonging to the Firmicutes phylum are numerically dominant in food waste (Fisgativa et al. 2018, Wu et al. 2018, Tang et al. 2023). Their metabolism, which generates volatile fatty acids, underpins the low pH of food waste, negatively influencing the efficiency of anaerobic digestion (Zhou et al. 2018). Interestingly, food waste harbours few representatives of one of the most diverse and abundant groups of microbes, Proteobacteria (Zhou et al. 2020a). Fresh vegetable and meat products are known to harbour Proteobacteria, such as *Serratia*, *Hafnia*, and *Pseudomonas* (Jackson et al. 2013, Remenant et al. 2015, Odeyemi et al. 2020), which play roles in food-borne disease and food spoilage (Jarvis et al. 2018). Proteobacteria are also commonly observed in food waste anaerobic digesters (Wu et al. 2016, Campanaro et al. 2020, Yadav et al. 2022).

Extracellular enzymes are responsible for biopolymer hydrolysis in anaerobic digestion (Menzel et al. 2020). The production of extracellular degradative enzymes by Proteobacteria has been intensively studied. Through microbiological and genomic studies, Gammaproteobacteria have been shown to possess the genetic ability to produce and secrete various extracellular hydrolytic enzymes (Mahmoudi et al. 2020). However, lactic acid bacteria have limited extracellular production, which is limited to the hydrolysis of polysaccharides and proteins (Wang et al. 2021).

In food waste environments characterized by high water activity, high nutrient availability, and low oxygen content, volatile fatty acid production by lactic acid bacteria lowers the pH. This results in the loss of Proteobacteria (Tang et al. 2023). The introduction of oxygen to maize and activated sludge prior to anaerobic digestion has been shown to increase the abundance of Proteobacteria (Ruan et al. 2019, Zhu et al. 2022). This phenomenon has not been investigated in the context of food waste pretreatment.

In this study, food waste samples were subjected to 3-day incubations with aeration and pH adjustment to neutral conditions, and the impacts on the volatile fatty acid concentration, extracellular enzyme production, microbial community composition, and subsequent anaerobic digestion were quantified. Through the understanding of the changes in volatile fatty acid and extracellular enzyme profile after pretreatment, the variations in this microbial-dependent activity can be linked to the alternation of pretreated food waste digestibility. Furthermore, the comparison between microbial communities in untreated and pretreated food waste could further indicate impactful microbes that emerged or dissipated during the post–pretreatment process. Pretreated food waste exhibited increased digestibility as observed in shortened anaerobic digestion time, along with altered VFA (Volatile fatty acid) profile, increased digestive enzyme activities, and growth of corresponding microbes.

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Table 1. Extracellular enzyme production by bacteria isolated from food waste.

Growing stages	Protease			Cellulase			Amylase			Lipase			Community relative abundance (%) (Tang et al. 2023)
Strain	Lag	Growth	Stationary	Lag	Growth	Stationary	Lag	Growth	Stationary	Lag	Growth	Stationary	
Lactiplantibacillus plantarum	-	-	+	-	-	+	-	-	-	-	-	-	5–30
Leuconostoc mesenteroides	-	-	+++	-	-	+	-	-	-	-	-	-	1–5
Bacillus halotolerans	-	-	-	-	-	-	-	-	+++	-	-	-	<0.1
Bacillus siamensis	-	-	+++	+	+++	-	-	-	+++	-	-	-	<0.1
Bacillus proteolyticus	-	-	-	+	+++	-	-	-	+++	-	-	-	<0.1
Bacillus aryabhattai	-	-	++	+	++	-	-	-	+++	-	-	-	<0.1
Arthrobacter koreensis	-	-	-	-	-	+	-	-	+++	-	-	+++	<0.1
Enterococcus faecalis	-	-	-	+	+	++	-	-	+++	-	+	++	<0.1
Bacillus amyloliquefaciens	-	-	-	-	-	-	-	-	-	-	-	-	<0.1
Bacillus altitudinis	-	-	-	+	++	++	-	-	-	-	-	-	<0.1
Bacillus velezensis	-	-	-	-	+	++	-	-	-	-	-	-	<0.1
Lysinibacillus fusiformis	-	-	-	++	+	++	-	-	-	-	-	+++	<0.1
Enterococcus gallinarum	-	-	-	++	++	+++	-	-	-	-	-	++	<0.1
Enterococcus faecium	-	-	-	++	+	++	-	-	-	-	-	-	<0.1
Arthrobacter luteolus	-	-	-	+++	++	-	-	-	-	-	-	-	<0.1
Bacillus stercoris	-	-	-	++	++	++	-	-	-	-	-	-	<0.1
Bacillus tequilensis	-	-	-	+++	+++	++	-	-	-	-	-	-	<0.1

Activity level is obtained by comparing within the enzyme column on a percentage basis with 100% being the highest observed activity value. (0%) No activity^{-}, (<33%) low activity^{+}, (<66%) medium activity^{++}, and (≥66%) high activity^{+++}.

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Table 2. Changes in pH, volatile fatty acid profile, and bacterial abundance in food waste in response to aeration and pH adjustment over 3 days.

	pH	Formate (mM)	Acetate (mM)	Propionate (mM)	Butyrate (mM)	Bacterial abundance (16S rRNA gene copy/g)
Prior to treatment	4.3 (±0.2)	58.7 (±75.6)	7.3 (±0.6)	21.0 (±4.2)	0 (±0)	2.5E + 9 (±9.8E + 8)
Untreated	4.5 (±0)*	14.7 (±1.2)	11.0 (±3.5)	14.0 (±1.4)*	0 (±0)	1.3E + 9 (±1.1E + 9)*
pH adjusted	6.1 (±0.3)†	15.0 (±1)	12.3 (±4.6)	69 (±29.7)†	136.7 (±78.5)†	2.3E + 10 (±2.8E + 10)
Aeration	5.0 (±0.7)	14.7 (±1.2)	8.7 (±1.2)	65.3 (±47.1)†	12.0 (±0)†	7.8E + 10 (±3.1E + 10)†
pH + aeration	6.7 (±0.1)†	14.7 (±1.2)	13.0 (±10.4)	103.3 (±44.6)†	41.5 (±3.5)†	3.0E + 10 (±4.2E + 10)

The values are averages from triplicate incubations (±standard deviation).
*t-test with $P < 0.05$ versus 'Prior to treatment'.
†t-test with $P < 0.05$ versus 'Untreated'.

Materials and methods

Food waste pretreatment

Food waste was obtained from a UNSW cafeteria, Sydney, Australia. The identifiable components in the food waste were bread, deli meat, leafy greens, tortillas, and tomatoes. Unwanted components such as napkins were manually discarded, and a food chopper was used to macerate the food waste into mince. Milli-Q water was added to food waste to make a 5% solid content food waste slurry. The food waste slurry (25 ml) was subjected to three types of pretreatments: pH adjustment (~pH 7); aeration, and aeration plus pH adjustment for 3 days at 32°C; and storing an untreated group and opening it to air at the same temperature. The pH was adjusted by adding phosphate buffer until the pH reached 7. The pH of the buffer was maintained above 6.5 for 3 days. Aeration was performed by shaking the samples, which were placed in 250 ml Erlenmeyer flasks and shaken at 160 rpm. Samples were taken on day 0 and day 3 and then stored at -20°C before analysis. The choice of a 3-day treatment period was based on both industry capability and the objective of minimizing VFA production (Charles et al. 2009).

Volatile fatty acid quantification

Acetate, formate, butyrate, and propionate were analysed with GC-FID (gas chromotography with downstream flame ionisation detector) (Carlström et al. 1965) using a DB-FFAP (nitroterephthalic-acid-modified polyethylene glycol) column (30 m × 0.32 mm × 0.25 mm, Agilent Technologies). An autosampler (Shimadzu AOC-5000 Plus) was used to directly inject headspace samples into the GC-FID instrument (GC-FID Shimadzu GC-2010). Sulfuric acid and ethanol were used to acidify the samples and standards. A mixing ratio of 5:2:2 (sample:ethanol:sulfuric acid) was used, followed by a 30-min incubation at 70°C. The GC test was set up at a temperature of 250°C and a pressure of 62 kPa. The gas flow rate was 25 ml/min in total, with 3 ml/min for the purge flow. Additionally, the sampling rate was 40 ms, with a sample incubation temperature of 40°C and a read time of 6 min.

Enzyme activities

Extracellular enzyme production for bacterial isolates and for food waste before and after pretreatment was quantified via published methods. The supernatant (filtered through a 0.22 µm filter) of pretreated food waste was used for the analysis. Samples for extracellular enzyme production at different growth stages were collected from isolates cultured for 1, 8 and 24 h at 30°C with shaking at 160 rpm in LB10 medium. *Arthrobacter koreensis* and *Bacillus stercoris* were selected for an alternative collection schedule because of their unique growth patterns. For these two strains, supernatant samples were collected at 3 h (instead of 8 h) for the lag phase.

The enzyme quantification protocol can be summarized as follows: the protease quantification method followed the protocol of Sigma–Aldrich and used casein as the substrate (Cupp-Enyard 2008). The cellulase activity was measured with carboxymethylcellulose in 0.05 M citrate buffer, and the absorbance was measured at 540 nm (Jasani et al. 2016). Dinitrosalicylic acid was used as the colouring agent. Amylase activity quantification was performed according to a published method (Oliveira et al. 2019). Like cellulase, dinitrosalicylic acid was used, and the absorbance was measured at 540 nm. The lipase activity quantification assay was performed according to a published protocol (Pinsirodom and Parkin 2001). Enzyme activities are measured as unit activity per ml of supernatant. All chemicals were purchased from Sigma–Aldrich unless specifically stated.

DNA isolation, sequencing, and quantitative PCR

DNA isolation was performed via the DNeasy PowerSoil Kit (Qiagen). The bacterial genetic copy abundance was quantified via qPCR (real-time polymerase chain reaction) via the primers 1048F and 1194R. The thermal cycling settings were as follows: 3 min of denaturation at 98°C, followed by 40 cycles of DNA segment replication at 90°C for 20 s and at 62°C for 50 s. The temperature was then decreased to 60°C and maintained for 10 s for fluorescent reads, followed by a temperature increase from 60 to 90°C in 0.5°C intervals for melt curve analysis.

Sequencing of the DNA was carried out by the Ramaciotti Center (UNSW) on the 16S rRNA gene V4 region via the primer set 515F&806R on the MiSeq V2 2 × 250 bp platform. Paired-end sequencing results were analysed via the QIIME II 2019.7 pipeline (Bolyen et al. 2019), and the sampling depth was set to 1600. The Dada2 plugin (Callahan et al. 2016) was used to denoise the data, with the trimming length determined via FastQC for quality assurance. The sequencing results have been submitted to the NCBI Sequence Read Archive (accession number PRJNA805020).

Biological methane potential assay

The Automatic Methane Potential Test System (AMPTS II from Bioprocess Control) runs 15 samples in parallel. Each bottle contained 150 ml of digestate and 150 ml of either food waste slurry or water as blank control. Five groups each containing biological triplicates include water, prior-to-treatment food waste, untreated food waste, aerated food waste, and aerated and pH-adjusted food waste. The water sample consisted of 150 ml of digestate and 150 ml of water. Prior to treatment, the food waste group was supplemented with 150 ml of food waste slurry with a solid content of 5% and 150 ml of digestate prior to treatment. The treated groups

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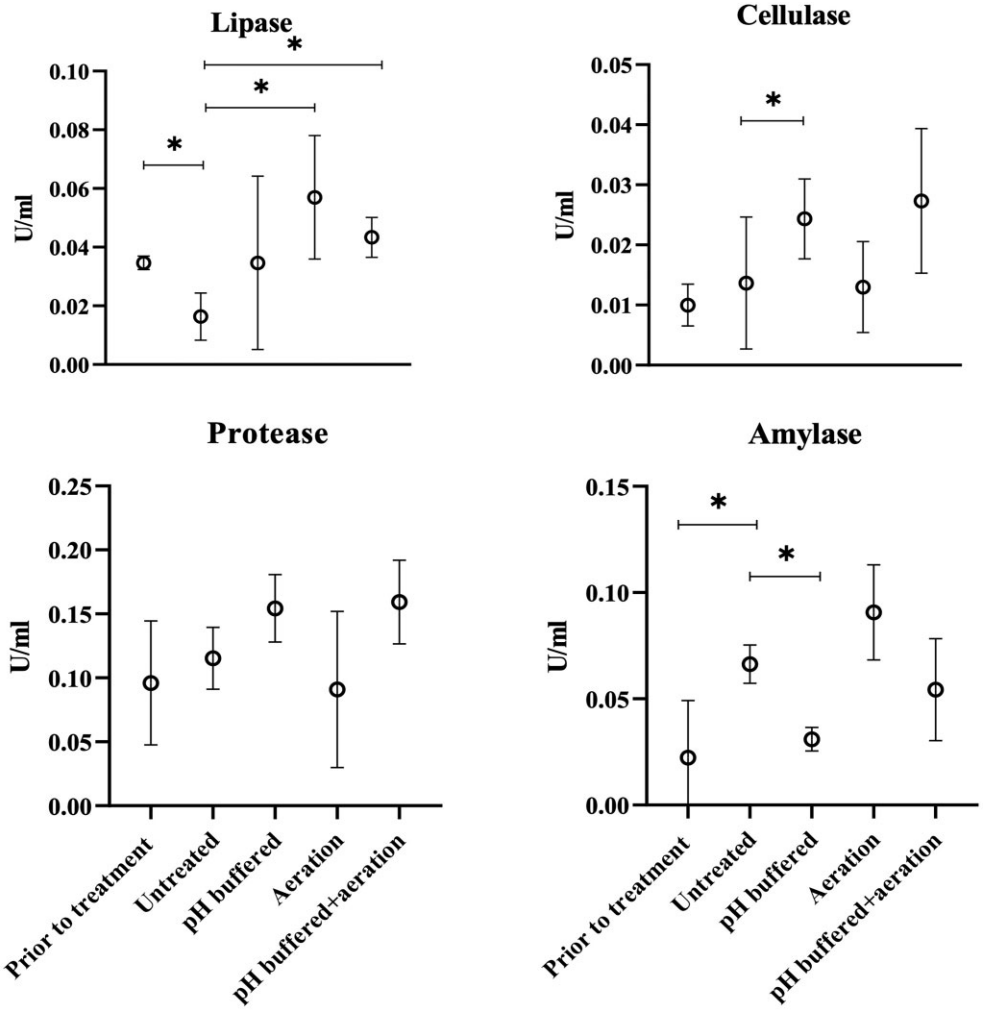


Figure 1. Extracellular enzyme activity under different pretreatment conditions. Enzyme activities were measured after 3 days of pretreatment and expressed in units of enzyme activity per ml of pretreated food waste supernatant. The data points are averages of triplicate incubations. The error bars represent the standard deviations. The horizontal bars and asterisks indicate significant differences between groups vertically below the two ends of the bar, with a *t*-test indicating $P < 0.05$.

contained 150 ml of digestate and 150 ml of treated food waste. The digestion was performed for 25 days. Methane production was measured by running the gas produced through a sodium hydroxide trap and an automatically registering gas trap. The data were collected automatically. All setup procedures were carried out as instructed by the manufacturer.

Statistical analysis method

Student's *t*-tests were used for statistical significance analysis in this study.

Results

Highly abundant bacteria in food waste display limited extracellular enzyme production

To assess the extracellular enzyme production of bacteria indigenous to food waste, 17 bacterial strains isolated from food waste were tested for their extracellular enzyme activities via culture supernatants from different growth stages. Table 1 shows that 24%–35% of the food waste isolates produced protease, amylase, and lipase activities exclusively during the late growth stage and early stationary phase. Cellulase activity was observed in 88% of the isolates whose production was observed throughout different growth stages. Isolates representing some of the most abundant bacteria in food waste samples (Lactobacillaceae species *Lactiplantibacillus plantarum* and *Leuconostoc mesenteriodes*) were only capable of producing protease and cellulase at relatively low amounts during the stationary phase, suggesting that the bacterial community indigenous to food waste has low potential for extracellular enzyme production.

Food waste pretreatment impacts volatile fatty acid production and bacterial abundance

To examine the impacts of pH neutralization and aeration on the indigenous microbial community of food waste, samples were incubated for 3 days with pH adjustment alone, aeration alone, or a combination of both. Table 2 shows that pH adjustment, aeration alone, or in combination resulted in significantly higher propionate concentrations (averaging 69.0, 65.3, and 103.3 mM compared with 14.0 mM in the untreated group and 21.0 mM in the prior-to-treatment group) and that pH adjustment with or without aeration resulted in significantly higher butyrate concentrations (averaging 106.7, 12.0, and 41.5 mM compared with 0 mM in both untreated and prior-to-treatment groups). Accom-

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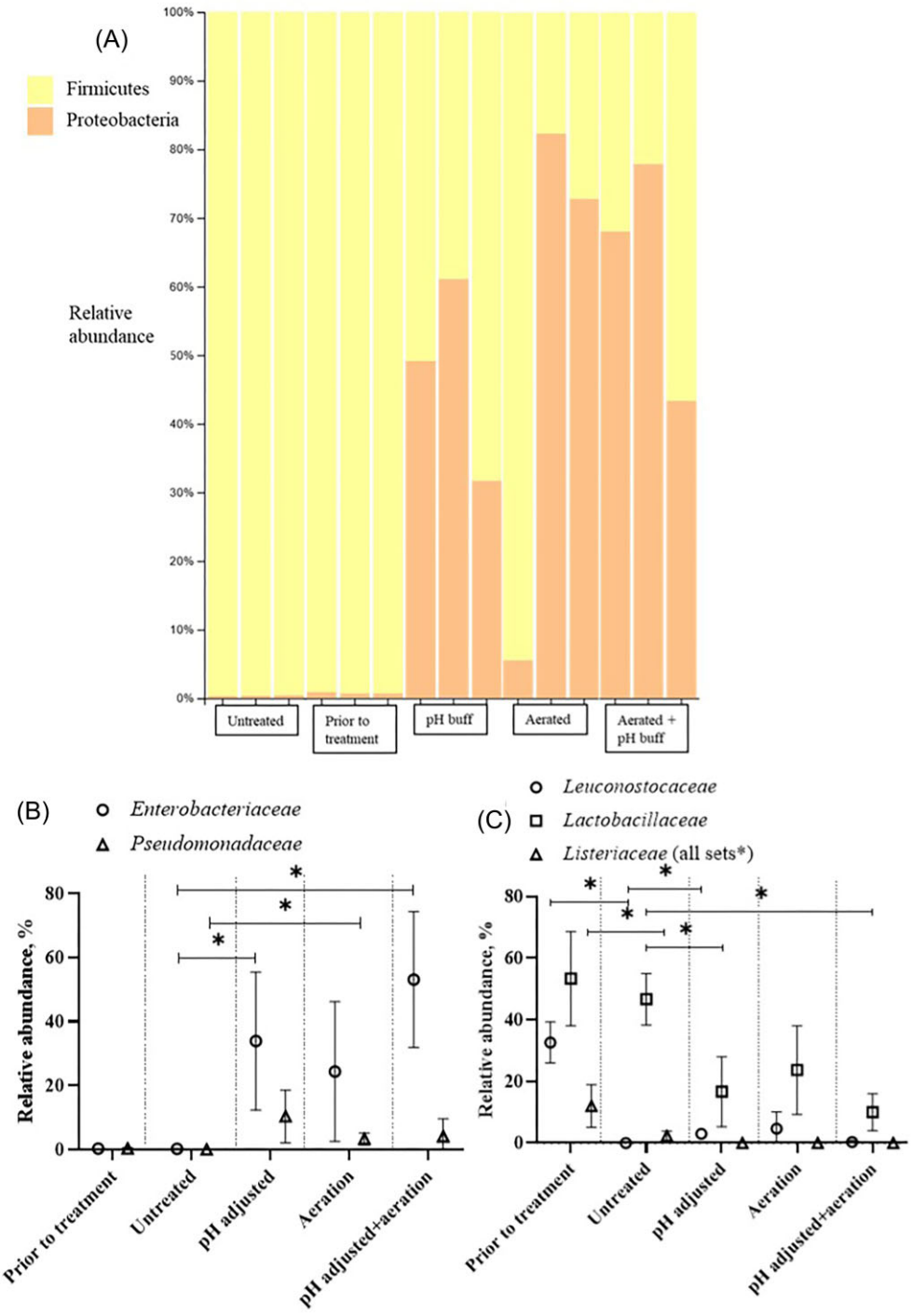


Figure 2. 16S rRNA gene relative abundance at the phylum level (A). Bacterial relative abundance (family-level) variation with respect to pretreatment regime (B and C). The three bars per group in (A) indicate results from three biological replicates per group. The horizontal bars and asterisks in (B) and (C) indicate significant differences between groups vertically below the end of the two bars, with a *t*-test indicating $P < 0.05$. The error bars represent the standard deviation. Each group included three biological replicates.

panying the increased propionate and butyrate acid production, formic acid production decreased in general regardless of treatment type (from 58.7 mM in the prior-to-treatment group to averaging 14.5 mM in the untreated, pH adjusted, aeration, and combine-treated groups). The bacterial abundance increase was significant in aerated incubations compared with that in untreated controls. Despite some increase observed in pH-adjusted or combined pH-adjusted and aerated incubations, statistical analysis deemed this increase insignificant when compared with untreated control. Additionally, aeration in combination with pH adjustment results in a greater pH than does pH adjustment alone ($P = 0.04$).

Food waste pretreatment impacts extracellular enzyme activity

Given the role of extracellular enzyme activity in food waste digestion, lipase, cellulase, protease, and amylase activities were quan-

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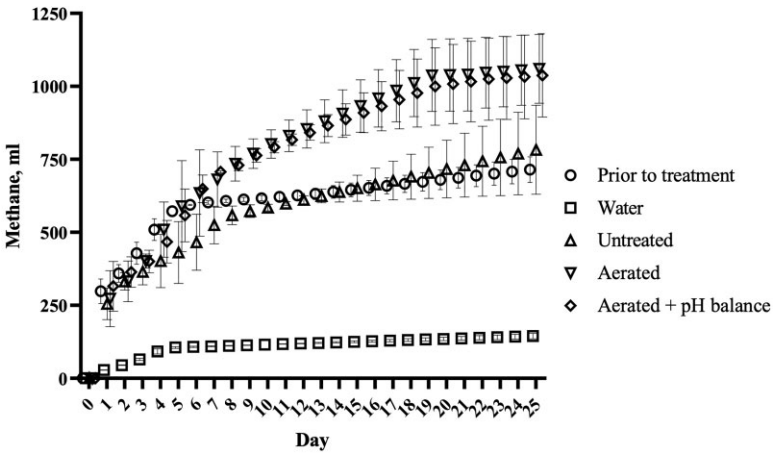


Figure 3. Methane production over 25 days of anaerobic digestion. The methane volume is the accumulated production of methane. The legend shows food waste under different treatment conditions, prior to treatment, pretreated or untreated, and a blank group using water as feedstock. The error bars indicate the standard deviation among three biological replicates.

tified before and after pretreatment. Aeration, with or without pH adjustment, increased lipase activity (Fig. 1). pH adjustment increased cellulase activity but decreased amylase activity. Protease activity was not affected by pH adjustment or aeration. All groups were compared with the prior-treatment group, and significant differences between the treated groups were also compared. Only the ones showing significant differences were indicated.

Pretreatment increased Proteobacteria and decreased Lactobacillaceae abundance

The impact of pretreatment on the microbial community composition after 3 days was examined by sequencing 16S rRNA genes in DNA extracts from food waste samples. The composition of the untreated samples was similar to that of food waste prior to pretreatment (Fig. 2 and Supplementary Fig. 1). pH adjustment, aeration, or a combination of both resulted in shifts in community composition on the basis of principal coordinate analysis (Supplementary Fig. 2). All the treatments significantly increased the proportion of Proteobacteria (Fig. 2). pH adjustment increased the relative abundance of Proteobacteria from 0.3% ± 0.1% in untreated samples to 44% ± 15% ($P = 0.02$). Aeration increased the relative abundance of Proteobacteria from 0.3% ± 0.1% in the untreated samples to 28.0% ± 23.0% ($P = 0.08$). The combined treatment increased the relative abundance of Proteobacteria from 0.3% ± 0.1% in the untreated samples to 57.0% ± 16.0% ($P = 0.01$). The increase in Proteobacteria was driven by the increase in Gammaproteobacteria, including an unclassified lineage within the Enterobacteriaceae family and *Pseudomonas psychrophile* in the Pseudomonadaceae family. A positive correlation ($R^2 = 0.92$) was observed between the relative abundance of Enterobacteriaceae and the propionate concentration (Supplementary Fig. 3). The relative abundances of Lactobacillaceae, Leuconostocaceae, and Listeriaceae decreased during pretreatment. pH was negatively correlated ($R^2 = 0.78$) with the relative abundance of Lactobacillaceae (Supplementary Fig. 3).

Pretreated food waste is more susceptible to anaerobic digestion

Biological methane potential tests were used to assess the impact of pretreatment on food waste anaerobic digestion (Fig. 3). Methane production from food waste prior to pretreatment was biphasic, with rapid methane production occurring over 5 days followed by a slow rate of methane production over the ensuing 20 days. Food waste pretreated for 3 days with aeration produced the most methane (averaging 1000 ml), and methane production was complete within 20 days. pH adjustment of aerated samples did not provide additional benefits to methane generation. The methane production profile of food waste samples that were incubated for 3 days without aeration or pH adjustment (untreated) was similar to that of food waste samples prior to pretreatment (relatively slow and ongoing after 25 days at roughly 750 ml).

Discussion

The development of pretreatment methods to enhance the anaerobic digestion of organic feedstocks such as food waste is a long-standing research endeavour. Traditionally, methods that harness the enzymatic activities of microorganisms indigenous to food waste, such as aeration, have been developed via a top-down approach, paying little attention to the composition of microbial communities and the relevant characteristics of specific microbial lineages. When microbiology has been investigated, a traditional cultivation-based approach has been taken, which is limited by widely recognized cultivation biases (Hassa et al. 2018).

We have taken a bottom-up approach to enhancing food waste digestion, viewing food waste and its microflora as an ecosystem, where emergent functions are dependent on microbial community composition. In a previous study, we characterized the indigenous microflora of food waste via contemporary DNA sequencing (Tang et al. 2023). This revealed that food waste communities are dominated by bacteria within the Firmicutes phylum, which are known for the production of organic acids that stave off competitors by lowering the pH (König and Fröhlich 2017). Moreover, bacteria within the phylum Proteobacteria, which is known for its role in food spoilage through extracellular enzyme production, were noticeably absent. Correlations between high Firmicutes abundance and low abundance of Proteobacteria have previously been observed in the beginning of anaerobic digestion (Wu et al. 2016). We therefore hypothesized that common pretreatments for enhanced anaerobic digestion, such as aeration and pH adjustment, are effective because they enhance the competitive fitness of Proteobacteria over Firmicutes in food waste ecosystems.

Macerated food waste samples were incubated for 3 days with and without aeration and pH neutralization. DNA sequencing

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confirmed the hypothesis that these pretreatments resulted in a significant shift in community composition from Firmicutes to Proteobacteria. More specifically, bacterial lineages within the Enterobacteriaceae and Pseudomonadaceae families increased in relative abundance, whereas lineages within the Leuconostocaceae, Lactobacillaceae, and Listeriaceae families decreased. Aeration resulted in an increase in pH and bacterial abundance, whereas pH neutralization alone or with aeration did not increase bacterial abundance. The shift towards increased relative abundance of Proteobacteria is therefore considered to be a function of pH, but the pH change driven by aeration is more conducive to growth within the 3-day incubation period. This finding can motivate the development of methods to increase proteobacterial abundance.

Aeration and pH neutralization affected the VFA profile of the food waste samples. Initially, formate followed by propionate was the most abundant VFA. Aeration and pH neutralization resulted in a decrease in formate concentration and an increase in propionate concentration. Proteobacteria may be associated with formate consumption and propionate production. According to various studies on VFA production, a pH of 5–6 favours propionic acid production, whereas a pH < 5 or > 6 favours butyric acid production (Sun et al. 2021). In both the aerated and/or pH-adjusted groups, the butyric acid concentration was greater than that in the aerated and untreated groups. High propionic acid production could also be linked to increased abundances of Clostridia, Betaproteobacteria, and Bacteroidetes (Zhou et al. 2020b).

However, within a complicated environment such as food waste, the concentration of VFAs will not be solely defined by production but will also be impacted by microbial consumption. Formate mainly comes from sugar fermentation by Enterobacterales (Leonhartsberger et al. 2002), whereas acetate can be fermented from propionate and butyrate (Zhang et al. 2009). Anaerobically, acetate can be used as a building block for butyrate (Duncan et al. 2004). Propionate can also be generated under aerobic conditions during food spoilage (Nychas et al. 2008). Acetate can be oxidized into longer carbon chain VFAs (Zhou et al. 2018). Longer carbon chain VFAs, such as butyrate, can be transformed into propionate and acetate and vice versa (Duncan et al. 2004). This could explain the observation that a decrease in butyrate accompanied an increase in propionate when the pH was adjusted and when the pH was adjusted and aerated. Formic acid can be used as an energy source for biological activities and is thus depleted after 3 days regardless of treatment. Gammaproteobacteria can metabolize formate (Zhou et al. 2020b), and propionate production is possible with an elevated abundance of Clostridium (Isipato et al. 2020). The lack of acetic acid could be the result of bacterial metabolism that ultimately hinders the dominance of lactic acid bacteria, which produce large amounts of acetic acid (Peinemann et al. 2020). Longer carbon chain VFAs may be derived from larger fatty acids being broken down into smaller VFAs during acidogenesis (Zhang et al. 2005, Liang et al. 2024). With more complex food waste substrates being transformed into various VFAs, pretreated food waste is more digestible during the anaerobic digestion process and has greater stability (Xu et al. 2021).

Enzymes produced by bacteria are essential for food waste digestion. Compared with the untreated samples, pretreated food waste resulted in greater digestive enzyme activity in biological methane potential tests over 25 days. The common hydraulic retention time for anaerobic food waste digestion is between 20 and 30 days (Zhang and Jahng 2012). The fact that methane production from pretreated food waste plateaus in <20 days with increased methane production compared with untreated food waste validates the effectiveness of pretreatment (Montalvo et al. 2016, Cheng et al. 2021).

In summary, this study revealed that aeration or pH neutralization pretreatment of food waste for 3 days resulted in a strong shift in the microbial community towards Proteobacteria. This was associated with increased extracellular enzyme activity and more efficient anaerobic digestion (reduced retention time). These findings suggest that manipulation of environmental parameters in food waste to promote the growth of Proteobacteria or artificial elevation of proteobacterial abundance through bioaugmentation can generate superior anaerobic digestion results.

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Author contributions

L.T. wrote the manuscript, acquired the data, and participated in the research design. M.M. led the research design and reviewed the manuscript. All the authors read and approved the manuscript.

Supplementary data

Supplementary data are available at *FEMSLE* online.

Conflict of interest: No conflicting interest needs to be declared by the authors.

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Data availability

The datasets generated during and/or analysed during the current study are available from the corresponding author upon reasonable request.

References

Bolyen E, Rideout JR, Dillon MR *et al.* Reproducible, interactive, scalable and extensible microbiome data science using QIIME 2. *Nat Biotechnol* 2019;**37**:852–7.

Callahan BJ, McMurdie PJ, Rosen MJ *et al.* DADA2: high-resolution sample inference from Illumina amplicon data. *Nat Methods* 2016;**13**:581–3.

Campanaro S, Treu L, Rodriguez-R LM *et al.* New insights from the biogas microbiome by comprehensive genome-resolved metagenomics of nearly 1600 species originating from multiple anaerobic digesters. *Biotechnol Biofuels* 2020;**13**:25.

Carlström G, Hallgren W, Pehrson B *et al.* Gas chromatographic determination of volatile fatty acids in aqueous media its use in bovine rumen fluid and in silage. *Acta Vet Scand* 1965;**6**:52–58.

Charles W, Walker L, Cord-Ruwisch R. Effect of pre-aeration and inoculum on the start-up of batch thermophilic anaerobic digestion of municipal solid waste. *Bioresour Technol* 2009;**100**:2329–35.

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Cheng L, Gao N, Quan C *et al.* Promoting the production of methane on the co-digestion of food waste and sewage sludge by aerobic pre-treatment. *Fuel* 2021;**292**:120197.
Cupp-Enyard C. Sigma's non-specific protease activity assay—casein as a substrate. *J Vis Exp* 2008;19:e899. https://doi.org/10.3791/899.
Duncan SH, Holtrop G, Lobley GE *et al.* Contribution of acetate to butyrate formation by human faecal bacteria. *Br J Nutr* 2004;**91**:915–23.
Fisgativa H, Tremier A, Saoudi M *et al.* Biochemical and microbial changes reveal how aerobic pre-treatment impacts anaerobic biodegradability of food waste. *Waste Manage (Oxford)* 2018;**80**:119–29.
Hassa J, Maus I, Off S *et al.* Metagenome, metatranscriptome, and metaproteome approaches unraveled compositions and functional relationships of microbial communities residing in biogas plants. *Appl Microbiol Biotechnol* 2018;**102**:5045–63.
Isipato M, Dessì P, Sánchez C *et al.* Propionate production by bioelectrochemically-assisted lactate fermentation and simultaneous CO_2 recycling. *Front Microbiol* 2020;**11**:599438.
Jackson CR, Randolph KC, Osborn SL *et al.* Culture dependent and independent analysis of bacterial communities associated with commercial salad leaf vegetables. *BMC Microbiol* 2013;**13**:274.
Jarvis KG, Daquigan N, White JR *et al.* Microbiomes associated with foods from plant and animal sources. *Front Microbiol* 2018;**9**:2540.
Jasani H, Umretiya N, Dharajiya D. Isolation, optimization and production of cellulase by *Aspergillus niger* from agricultural waste. *J Pure Appl Microbiol* 2016;**10**:1159–66.
König H, Fröhlich J. Lactic acid bacteria. In: König H, Unden G, Fröhlich J (eds), *Biology of Microorganisms on Grapes, in Must and in Wine*. Cham: Springer International Publishing, 2017;3–41.
Leonhartsberger S, Korsa I, Böck A. The molecular biology of formate metabolism in enterobacteria. *J Mol Microbiol Biotechnol* 2002;**4**:269–76.
Liang Y, Chen Y, Chen C *et al.* Sequential bio-treatment of ammonia-rich wastewater from Chinese medicine residue utilization: regulation of dissolved oxygen. *Bioresour Technol* 2024;**406**:131041.
Mahmoudi N, Hagen SM, Hazen TC *et al.* Patterns in extracellular enzyme activity and microbial diversity in deep-sea Mediterranean sediments. *Deep Sea Res Part I* 2020;**158**:103231.
Menzel T, Neubauer P, Junne S. Role of microbial hydrolysis in anaerobic digestion. *Energies* 2020;**13**:5555.
Montalvo S, Huiliñir C, Ojeda F *et al.* Microaerobic pretreatment of sewage sludge: effect of air flow rate, pretreatment time and temperature on the aerobic process and methane generation. *Int Biodeterior Biodegrad* 2016;**110**:1–7.
Nychas G-JE, Skandamis PN, Tassou CC *et al.* Meat spoilage during distribution. *Meat Sci* 2008;**78**:77–89.
Odeyemi OA, Alegbeleye OO, Strateva M *et al.* Understanding spoilage microbial community and spoilage mechanisms in foods of animal origin. *Comp Rev Food Sci Food Safe* 2020;**19**:311–31.
Oliveira HM, Pinheiro AQ, Fonseca AJM *et al.* Flexible and expeditious assay for quantitative monitoring of alpha-amylase and amyloglucosidase activities. *MethodsX* 2019;**6**:246.
Peinemann JC, Rhee C, Shin SG *et al.* Non-sterile fermentation of food waste with indigenous consortium and yeast—effects on microbial community and product spectrum. *Bioresour Technol* 2020;**306**:123175.
Pinsirodom P, Parkin KL. Lipase assays. *Curr Protoc Food Anal Chem* 2001;00:C3.1.1–C3.1.13.
Remenant B, Jaffrès E, Dousset X *et al.* Bacterial spoilers of food: behavior, fitness and functional properties. *Food Microbiol* 2015;**45**:45–53.
Ruan D, Zhou Z, Pang H *et al.* Enhancing methane production of anaerobic sludge digestion by microaeration: enzyme activity stimulation, semi-continuous reactor validation and microbial community analysis. *Bioresour Technol* 2019;**289**:121643.
Suhartini S, Rohma NA,Elviliana *et al.* Food waste to bioenergy: current status and role in future circular economies in Indonesia. *Energ Ecol Environ* 2022;**7**:297–339.
Sun J, Zhang L, Loh K-C. Review and perspectives of enhanced volatile fatty acids production from acidogenic fermentation of lignocellulosic biomass wastes. *Bioresour Bioprocess* 2021;**8**:1–21.
Tang L, O'Dwyer J, Kimyon Ö *et al.* Microbial community composition of food waste before anaerobic digestion. *Sci Rep* 2023;**13**:12703.
Wang Y, Wu J, Lv M *et al.* Metabolism characteristics of lactic acid bacteria and the expanding applications in food industry. *Front Bioeng Biotechnol* 2021;**9**:612285.
Wu B, Wang X, Deng Y-Y *et al.* Adaption of microbial community during the start-up stage of a thermophilic anaerobic digester treating food waste. *Biosci Biotechnol Biochem* 2016;**80**:2025–32.
Wu S, Xu S, Chen X *et al.* Bacterial communities changes during food waste spoilage. *Sci Rep* 2018;**8**:8220.
Xu H, Li Y, Hua D *et al.* Effect of microaerobic microbial pretreatment on anaerobic digestion of a lignocellulosic substrate under controlled pH conditions. *Bioresour Technol* 2021;**328**:124852.
Yadav M, Joshi C, Paritosh K *et al.* Reprint of organic waste conversion through anaerobic digestion: a critical insight into the metabolic pathways and microbial interactions. *Metab Eng* 2022;**71**:62–76.
Zhang H, DiBaise JK, Zuccolo A *et al.* Human gut microbiota in obesity and after gastric bypass. *Proc Natl Acad Sci USA* 2009;**106**:2365–70.
Zhang L, Jahng D. Long-term anaerobic digestion of food waste stabilized by trace elements. *Waste Manage (Oxford)* 2012;**32**:1509–15.
Zhang B, Zhang LL, Zhang SC *et al.* The influence of pH on hydrolysis and acidogenesis of kitchen wastes in two-phase anaerobic digestion. *Environ Technol* 2005;**26**:329–40.
Zhou Z, Liu Y, Pan J *et al.* Gammaproteobacteria mediating utilization of methyl-, sulfur- and petroleum organic compounds in deep ocean hydrothermal plumes. *ISME J* 2020;**14**:3136–48.
Zhou XL, Ouyang Z, Zhang XL *et al.* Effects of a high-dose *Saccharomyces cerevisiae* inoculum alone or in combination with *Lactobacillus plantarum* on the nutritional composition and fermentation traits of maize silage. *Anim Prod Sci* 2020;**60**:833–42.
Zhou M, Yan B, Wong JWC *et al.* Enhanced volatile fatty acids production from anaerobic fermentation of food waste: a mini-review focusing on acidogenic metabolic pathways. *Bioresour Technol* 2018;**248**:68–78.
Zhu R, Wang D-H, Zheng Y *et al.* Understanding the mechanisms behind micro-aeration to enhance anaerobic digestion of corn straw. *Fuel* 2022;**318**:123604.

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